7/*Given an unsorted array of integers, design an algorithm and implement a program to sort this array using selection sort. Your program should also find number of comparisons and number of swaps required.

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*/

```
#include<iostream>
#include<vector>
using namespace std;
void selectionsort(vector<int>&vec){
    int comparison=0;
    int swaps=0;
    for(int i=0;i<vec.size()-1;i++){

        int j=i+1;
        int min_index=i;

        while(j<vec.size()){
            comparison++;
            if(vec[j]<vec[min_index]){
                min_index=j;
            }
            j++;
        }
        if(i!=min_index){
            swap(vec[i],vec[min_index]);
            swaps++;
        }

    }
    cout<<"comparison: "<<comparison<<endl;
    cout<<"swaps: "<<swaps<<endl;
}

int main()
{
    vector<int> vec;
```

```
    int choice, num;

    while (1)
    {
        cout << "Enter the number: ";
        cin >> num;
        vec.push_back(num);

        cout << "Want to continue? (1 for yes, 2 for no): ";
        cin >> choice;

        if (choice == 2)
            break;
    }

    selectionsort(vec);
    cout << "Sorted array: ";
    for (int x : vec)
        cout << x << " ";
}
```

OUTPUT:

```
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> g++
selection_sort.cpp
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> ./a
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 2
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 2

comparison: 3
swaps: 0
Sorted array: 1 2 3
--------------------------------------------------------------------------------
Enter the number: 5
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 2

comparison: 3
swaps: 1
Sorted array: 1 3 5
--------------------------------------------------------------------------------
Enter the number: 64
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 25
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 12
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 22
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 11
Want to continue? (1 for yes, 2 for no): 2

comparison: 10
swaps: 2
Sorted array: 11 12 22 25 64
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem>
```

8./* Given an unsorted array of integers, design an algorithm and a program to sort the array using insertion sort. Your program should be able to find number of comparisons and shifts (shifts - total number of times the array elements are shifted from their place) required for sorting the array.

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```
 */
#include <iostream>
#include <vector>
using namespace std;

void insertion_sort(vector<int>& vec)
{
    int comparison = 0;
    int shift = 0;

    for (int i = 1; i < vec.size(); i++)
    {
        int temp = vec[i];
        int j = i - 1;

        while (j >= 0)
        {
            comparison++;

            if (vec[j] > temp)
            {
                vec[j + 1] = vec[j];
                shift++;
                j--;
            }
            else
            {
                break;
            }
        }
        vec[j + 1] = temp;
    }
```

```
    cout << "Comparisons: " << comparison << endl;
    cout << "Shifts: " << shift << endl;
}

int main()
{
    vector<int> vec;
    int choice, num;

    while (1)
    {
        cout << "Enter the number: ";
        cin >> num;
        vec.push_back(num);

        cout << "Want to continue? (1 for yes, 2 for no): ";
        cin >> choice;

        if (choice == 2)
            break;
    }

    insertion_sort(vec);

    cout << "Sorted array: ";
    for (int x : vec)
        cout << x << " ";
}
```

OUTPUT:

```
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> g++
insertion_sort.cpp
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> ./a
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 2
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 2

Comparisons: 2
Shifts: 0
Sorted array: 1 2 3
--------------------------------------------------------------------------------
Enter the number: 5
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 2

Comparisons: 3
Shifts: 3
Sorted array: 1 3 5
--------------------------------------------------------------------------------
Enter the number: 64
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 25
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 12
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 22
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 11
Want to continue? (1 for yes, 2 for no): 2

Comparisons: 9
Shifts: 7
Sorted array: 11 12 22 25 64
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem>
```

9./* Given an unsorted array of positive integers, design an algorithm and implement it using a program to find whether there are any duplicate elements in the array or not.
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Roll no – 39

*/

```
#include<iostream>
#include<vector>
#include<unordered_map>
using namespace std;
void count_duplicates(vector<int>&vec){
   unordered_map<int,int>mp;
   for(int i=0;i<vec.size();i++){
      mp[vec[i]]++;
   }
   for(auto it:mp){
      if(it.second>1){
         cout<<"duplicates are present!"<<endl;
         return;
      }
   }
   cout<<"no duplicates are present!"<<endl;
}
int main(){
     vector<int> vec;
   int choice, num;

   while (1)
   {
      cout << "Enter the number: ";
      cin >> num;
      vec.push_back(num);

      cout << "Want to continue? (1 for yes, 2 for no): ";
      cin >> choice;

      if (choice == 2)
         break;
   }
```

```
    count_duplicates(vec);
}
```

OUTPUT:

```
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> g++
count_duplicates.cpp
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> ./a
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 2
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 2

no duplicates are present!
--------------------------------------------------------------------------------
Enter the number: 5
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 5
Want to continue? (1 for yes, 2 for no): 2

duplicates are present!
--------------------------------------------------------------------------------
Enter the number: 10
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 20
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 10
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 30
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 20
Want to continue? (1 for yes, 2 for no): 2

duplicates are present!
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem>
```

10./* Given an unsorted array of integers, design an algorithm and implement it using a program to sort an array of elements by dividing the array into two subarrays and combining these subarrays after sorting each one of them. Your program should also find number of comparisons and inversions during sorting the array.
Name-Parth Nailwal

Course-Btech Cse

Section- B1

Roll no – 39

*/

```
#include <iostream>
#include <vector>
using namespace std;

void merge(vector<int>& vec, int left, int mid, int right) {
    int i = left;
    int j = mid + 1;
    int k = 0;

    vector<int> temp(right - left + 1);


    while (i <= mid && j <= right) {
        if (vec[i] <= vec[j]) {
            temp[k++] = vec[i++];
        } else {
            temp[k++] = vec[j++];
        }
    }


    while (i <= mid) {
        temp[k++] = vec[i++];
    }


    while (j <= right) {
        temp[k++] = vec[j++];
    }
```

```
    for (int x = 0; x < temp.size(); x++) {
        vec[left + x] = temp[x];
    }
}

void merge_sort(vector<int>& vec, int left, int right) {
    if (left < right) {
        int mid = left + (right - left) / 2;
        merge_sort(vec, left, mid);
        merge_sort(vec, mid + 1, right);
        merge(vec, left, mid, right);
    }
}

int main() {
    vector<int> vec;
    int choice, num;

    while (1) {
        cout << "Enter the number: ";
        cin >> num;
        vec.push_back(num);

        cout << "Want to continue? (1 for yes, 2 for no): ";
        cin >> choice;

        if (choice == 2)
            break;
    }

    merge_sort(vec, 0, vec.size() - 1);

    cout << "Sorted array: ";
    for (int x : vec)
        cout << x << " ";
}
```

OUTPUT:

```
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> g++
merge_sort.cpp
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem> ./a
Enter the number: 1
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 2
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 2

Sorted array: 1 2 3
--------------------------------------------------------------------------------
Enter the number: 9
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 6
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 2

Sorted array: 3 6 9
--------------------------------------------------------------------------------
Enter the number: 38
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 27
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 43
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 3
Want to continue? (1 for yes, 2 for no): 1
Enter the number: 9
Want to continue? (1 for yes, 2 for no): 2

Sorted array: 3 9 27 38 43
PS C:\Batch2024-2028\CS4(B1)\Parth Nailwal 39\DAA 4th sem>
```